

Dates-as-Data for Inscriptions

Using SPA with Latin epigraphic databases to investigate changing complexity in the Western Roman Empire — RAC-TRAC 2026, TRAC7, Aarhus

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1 · The dates-as-data tradition, and a persistent puzzle

- 40 years of “dates as data” in archaeology (Rick 1987; Williams 2012; Timpson et al. 2014; Crema & Bevan 2021)
- LIRE v3.0: **180,609 inscriptions**, 50 BC – AD 350 (Kaše et al. 2023)
- **The recurring objection:** counts reflect *cultural enthusiasm to inscribe* — the **epi-graphic habit** — more than anything else (MacMullen 1982; Hanson 2021)
- **Our question:** how much demographic signal survives once we account for editorial distortion?

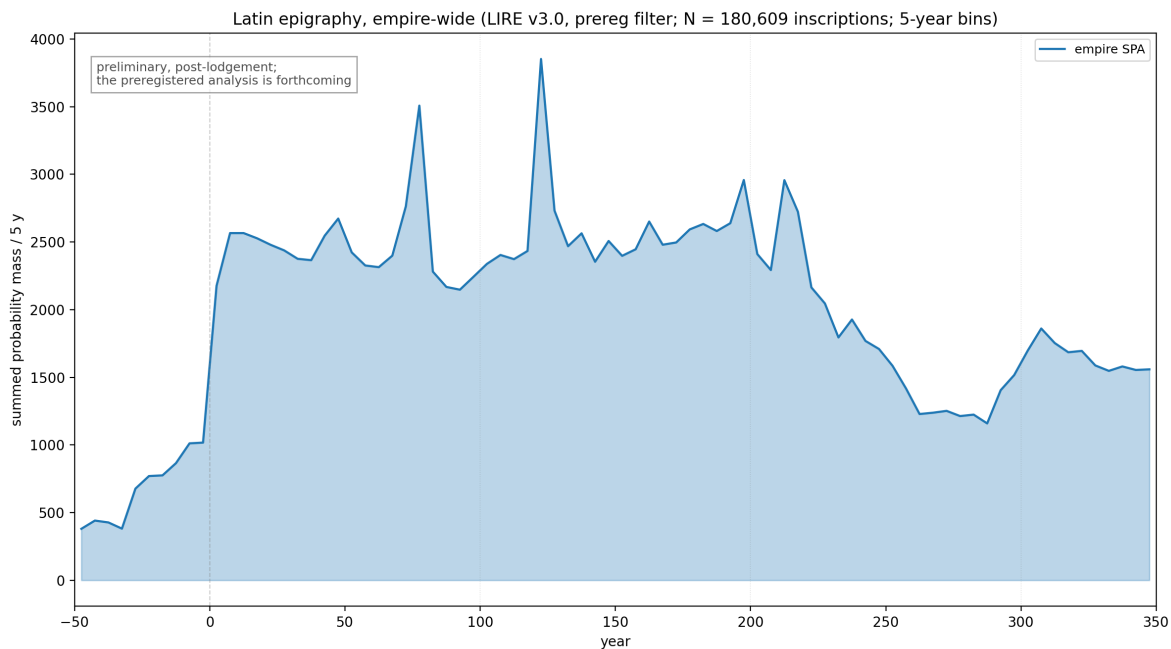
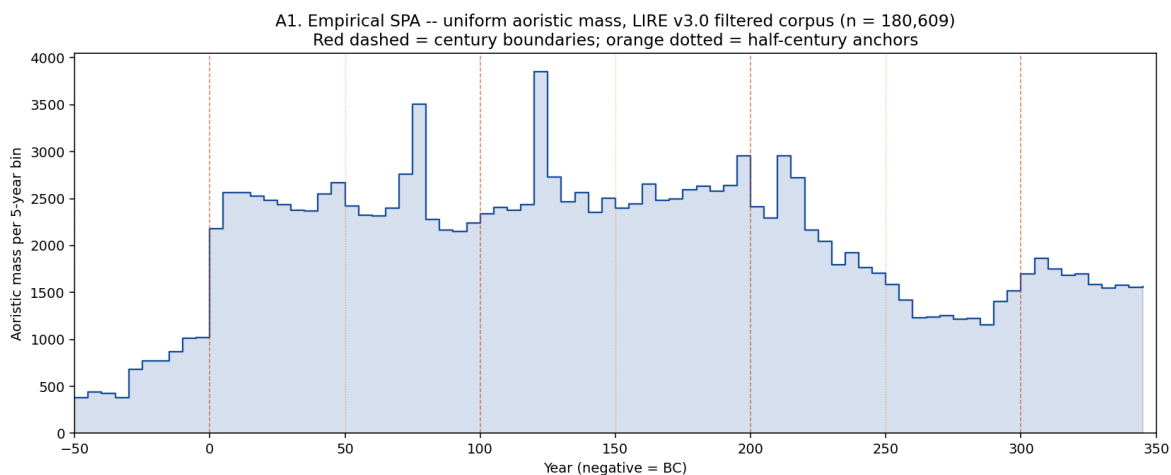
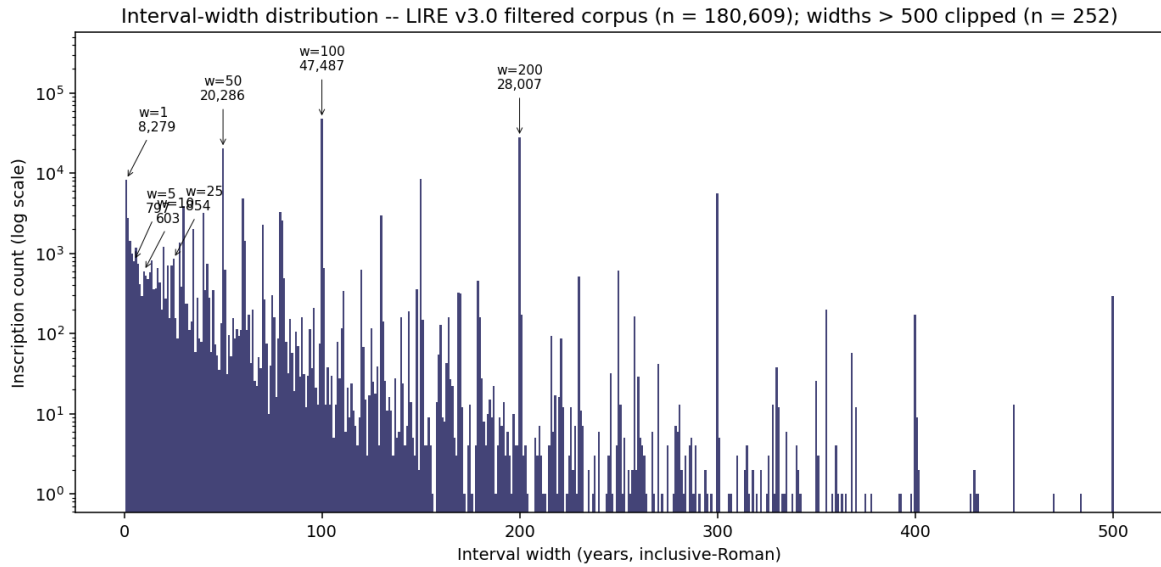


Figure 1: Empire-wide SPA of LIRE v3.0 under the preregistered 50 BC – AD 350 date-window filter (180,609 inscriptions; 5-year bins; uniform aoristic mass). The plateau-step structure at century boundaries and narrow spikes at AD 77.5 / 122.5 / 212.5 are the empirical signature of editorial dating conventions.

2 · Editorial distortion has a measurable signature



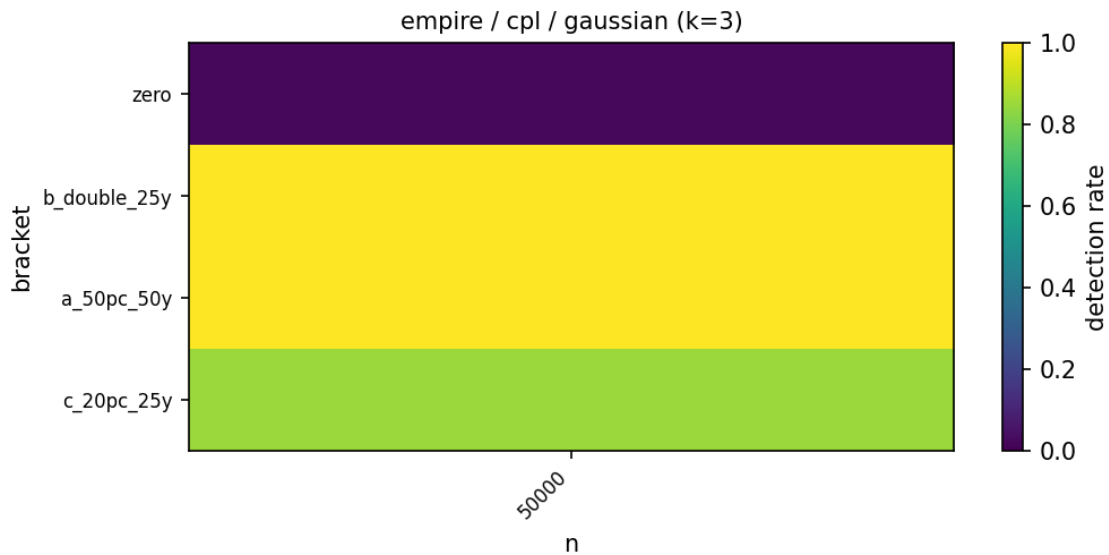
- **54.5 %** of `not_before` end in 01; **53.0 %** of `not_after` end in 00
- Largest step in the SPA: **+1,159 at the 1 BC / AD 1 boundary**
- Narrow spikes at AD 77.5 (Flavian), 122.5 (Hadrianic), 212.5 (Severan)



Reading the signature:

- Wide templates → editorial encoding artefact
- Narrow spikes → real ancient clustering (sustain under high-precision filter)

3 · What we can resolve: Phase 1 reachability



Phase 1 asks: at each analysis level, how many inscriptions does a unit need before deviation-detection reliably sees a known effect?

For the binding 50 %-over-50-y bracket (Gaussian-tapered, conservative):

- **Empire:** reachable at $n = 50,000$
- **Province / Urban area:** min n **1,549** (exp / Gauss; range 1,385 – 1,938 across nulls)
- **96 H1-reachable cells** across empire / province / urban-area

These thresholds **gate** the per-subset deviation-detection analyses; locked at OSF lodgement.

4 · What the data look like at multiple scales

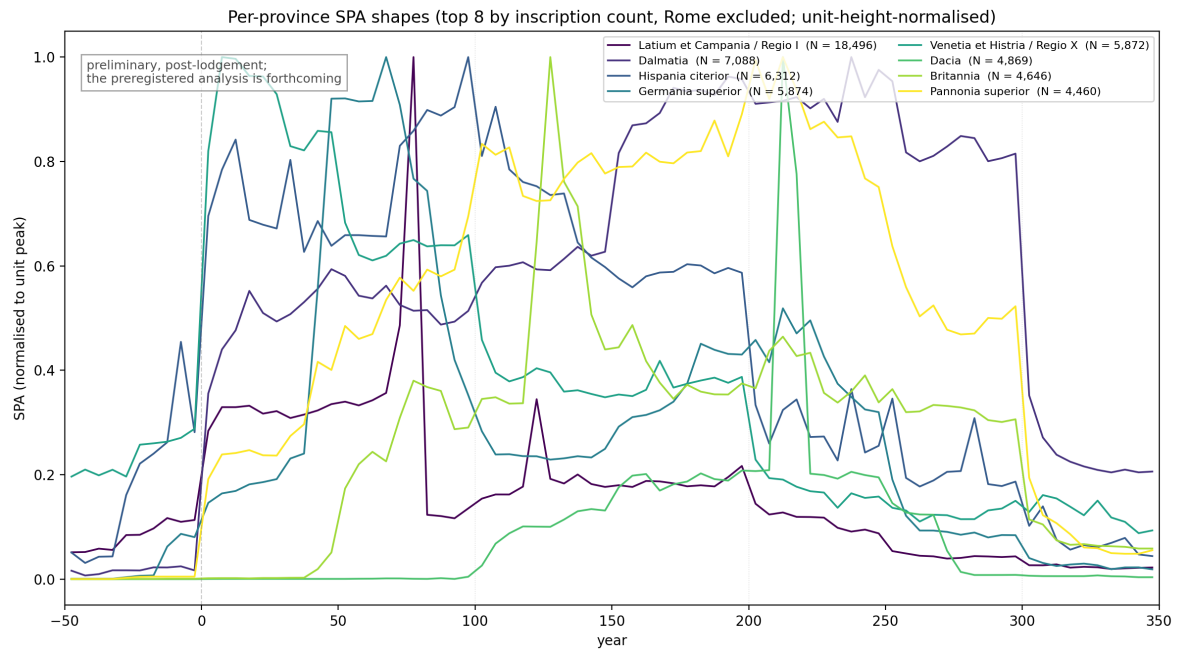


Figure 2: Top 8 provinces by inscription count, Rome-excluded; each scaled to unit peak height. Shape *and* amplitude differ.

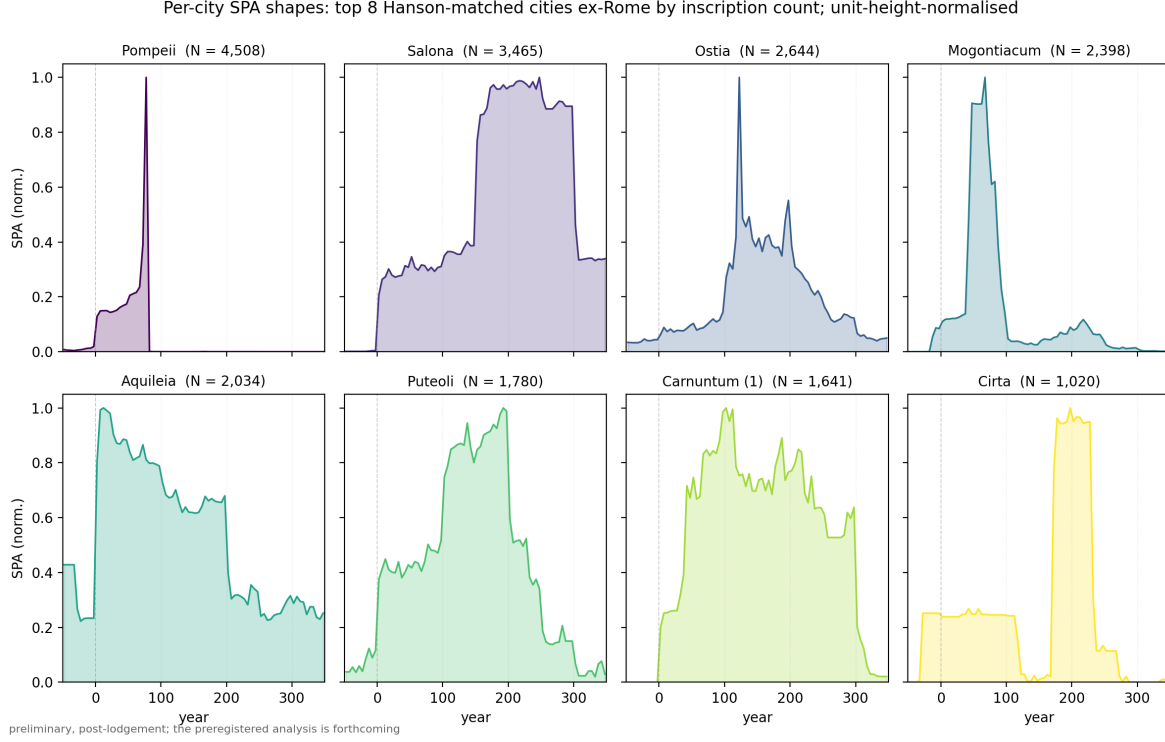


Figure 3: Top 8 Hanson-matched cities (ex-Rome); each scaled to unit peak. Pompeii cuts off at AD 79 (eruption); Ciria peaks 2nd–3rd c. (Antonine boom).

- Editorial distortion (plateau-steps) visible; **mixture correction is the preregistered Phase 2 work** (next slide)
- **Pompeii cuts off at AD 79** — smoking-gun validity check

5 · The preregistered fix: Bayesian mixture decomposition

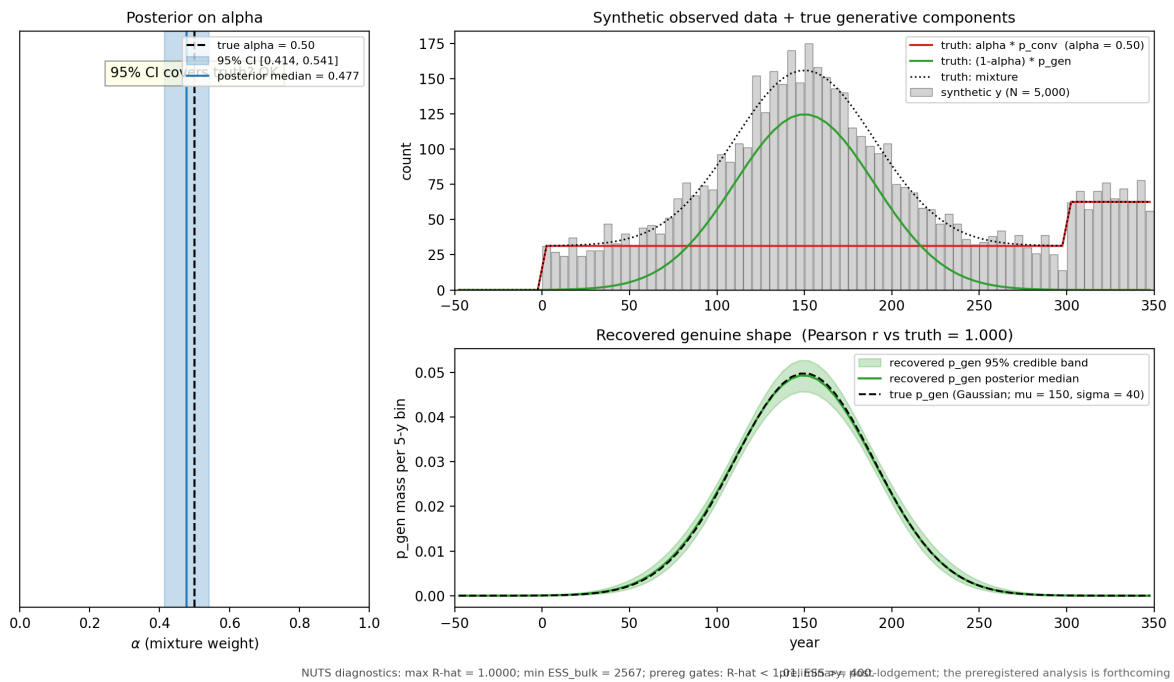
Model (preregistered):

$$y_t \sim \text{Multinomial}(N, \alpha \cdot p_t^{\text{conv}} + (1 - \alpha) \cdot p_t^{\text{gen}})$$

- p^{conv} = template-slab dictionary (century / half-century / reign-interval)
- p^{gen} = smoothed “real ancient” SPA
- α = empire-level mixing weight

Validation: preregistered recovery-simulation grid; supplementary Dirichlet-multinomial + rescaled NegBin fits reported alongside.

Synthetic recovery of the preregistered Bayesian mixture model (one cell)



Synthetic cell, $N = 5,000$, true $\alpha = 0.50$. **posterior covers truth** (median 0.477, 95 % CI [0.414, 0.541]); **shape Pearson $r = 1.000$** ; all prereg validation gates pass. *Schematic only; full grid post-talk.*

6 · Preliminary scaling: does Hanson population predict inscription count?

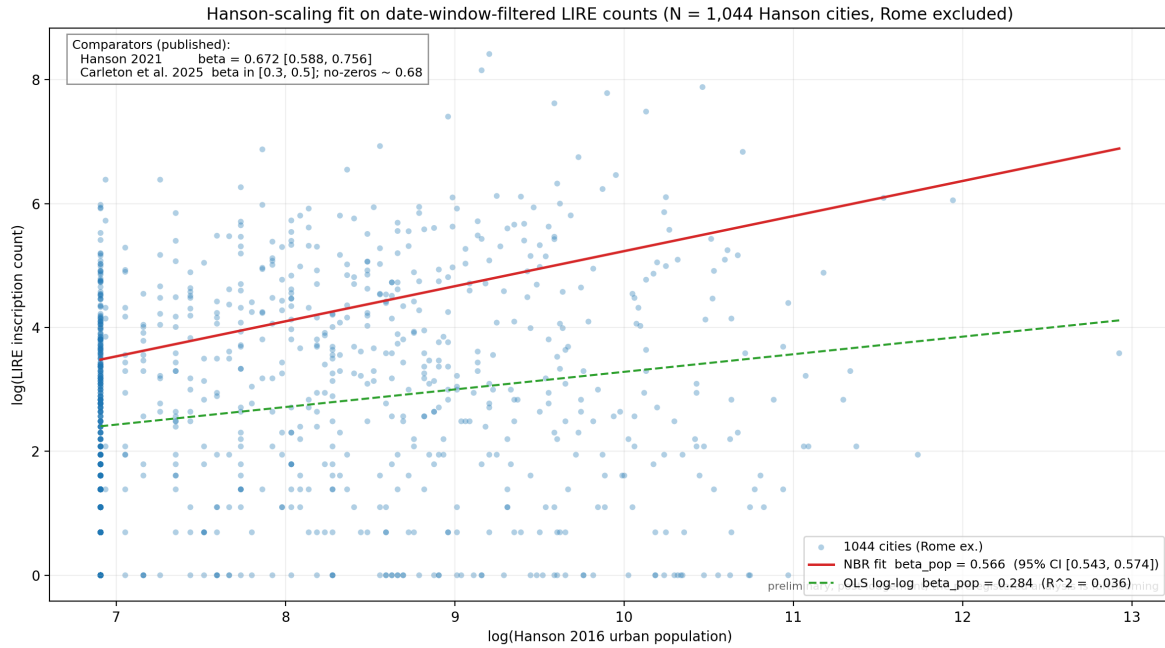


Figure 4: Log-log scatter of inscription count vs Hanson 2016 urban-area population (1,044 Hanson-matched cities, Rome excluded). Frequentist NBR fit ($\beta = 0.566$) and OLS log-log fit ($\beta = 0.284$) shown.

Preliminary frequentist result (date-window-filtered counts; Rome excluded):

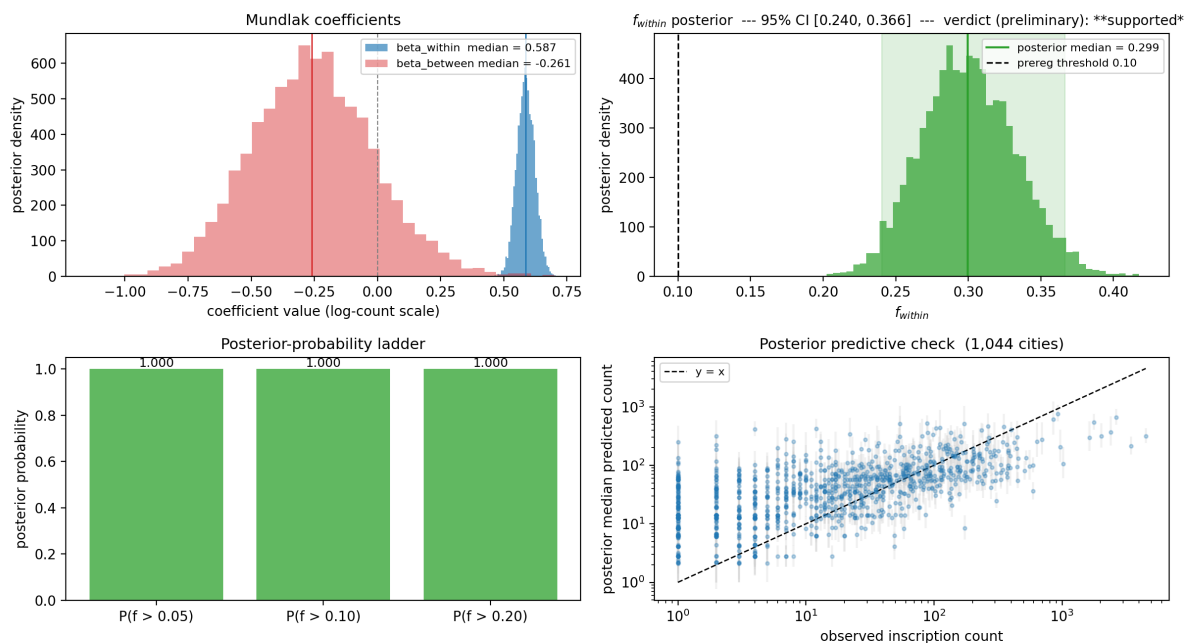
Statistic	Value	Comparator
NBR (preliminary)	0.566	Hanson 2021: $\beta = 0.672$ (95 % CI [0.588, 0.756])
Bootstrap 95 % CI	[0.543, 0.574]	Carleton et al. 2025 (elite-honorific): 0.3–0.5; no-zeros ~ 0.68
OLS log-log	0.284 ($R^2 = 0.04$)	OLS is the Hanson-direct comparator, but is dampened here by the low-count tail
n (cities)	1,044	Hanson 2021: 554 sites

Claim: population accounts for *some* of the cross-city variation in inscription count — the habit-only critique cannot be the whole story.

Caveats: - Frequentist NBR; the **binding** analysis is the Bayesian within-between Mundlak NBR on the next slide - Row-resample bootstrap underestimates between-city heterogeneity vs the Mundlak posterior - Cumulative (not peak) counts vs Hanson's max-theoretical populations - This is **preliminary, post-lodgement**; not a confirmatory test

6b · The preregistered binding analysis: within-province population effect

H3a Bayesian within-between (Mundlak) NBR --- preliminary post-lodgement results



NUTS: chains=4, warmup=3000, draws=2000 --- max R-hat = 1.0000 (gate < 1.01); min ESS_bulk = 8612 (gate=100); all parameters converged. The preregistered analysis is forthcoming

Mundlak NBR on 1,044 cities ex-Rome:

- $\beta_{\text{within}} = 0.587$
- $\beta_{\text{between}} = -0.26$ (CI straddles 0; not separately identifiable)
- $f_{\text{within}} = 0.299$, 95 % CI [0.240, 0.366]
- Verdict (3-way): supported

~ 30 % of city-to-city variation → within-province population. The other ~ 70 % = habit + convention + survival + admin structure.

7 · Where this is heading, and what we'd love feedback on

Lodged on OSF 2026-05-20 — osf.io/uycs6 (*embargoed pending review decision*):

- **Phase 2:** Bayesian mixture validation (recovery grid)
- **Phase 3 / H3a:** Mundlak NBR + `f_within` verdict (today's headline result, preliminary)
- **H3b:** deviation-detection at Antonine plague + 3rd-c. Crisis
- **H3c:** provincial-capital residuals + Moran's I (Hanson 2021 replication)
- 13 named sensitivity / exploratory analyses

Public repo at the lodgement tag (link in footer).

Questions for this room:

1. **For the LIRE / SDAM team:** does the editorial-template decomposition match how dates are assigned in EDH / EDCS / LIRE?
 2. Mundlak attributes ~ **30 %** of city-to-city variation to within-province population. **Which sub-mechanisms** drive the other ~ 70 % — and how would we distinguish them?
 3. **Negative-control subgroups** (habit-dominant) or **independent demographic anchors** we should consult?
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Backup slides (not in main 12-min flow)

B1 · Hanson's residual findings

If asked about replicating Hanson 2021's specific findings: - Provincial capitals over-produce inscriptions (mean residual 0.43 vs ~0.06 for *coloniae* / *municipia*) - Moran's I = 0.046 on residuals (significant clustering) - Moran's I = -0.006 on raw counts (no clustering — the clustering is in the residuals)

Our H3c is a preregistered replication of both findings.

B2 · Why both frequentist and Bayesian on the deck?

- The **frequentist NBR** (slide 6) is the direct comparator to Hanson 2021 ($\beta = 0.672$) and Carleton et al. 2025 ($\beta = 0.3 - 0.5$; no-zeros $\beta = 0.68$); it's an empire-wide scaling coefficient
- The **Bayesian within-between Mundlak NBR** (slide 6b) is the *preregistered binding analysis* — it isolates the *within-province* population effect, orthogonal to province membership, and yields a posterior on the variance-fraction estimand f_{within}
- The two specifications answer related but distinct questions; the Mundlak $\beta_{\text{within}} = 0.587$ sits inside the bootstrap CI of the empire-wide NBR $\beta = 0.566$ $[0.543, 0.574]$ — qualitatively consistent
- Sample sizes differ slightly between the two: 1,044 Hanson-matched cities ex-Rome in both; the Mundlak fit further partitions by 56 provinces for the random-intercept structure

B3 · Why the 50 BC – AD 350 envelope?

- Constrains the data-attribution artefact (late-Republic dating is sparse; post-AD 350 is Late Antique with different epigraphic conventions and different cultural-political context)
- Aligned with Hanson 2016's reference period (100 BC – AD 300, with our extension to AD 350)
- Roughly the imperial-era period during which urban populations and inscription production overlap

B4 · Why exclude Rome?

- Rome contributes 65,435 inscriptions to the filtered corpus — 36.2 % of the total
- It dominates the scaling fit as a single outlier
- Hanson 2021 also excludes Rome from scaling (Table 7.3 caption)
- Reported transparently; not tested as a sensitivity (per prereg §9)

B5 · How the mixture model handles year-precise inscriptions

- Year-precise inscriptions ($[t, t]$ encodings; consular or imperial-titulature dating) are *not* part of the convention component
- They remain in the genuine SPA as real ancient anchors
- The convention component is built from the *wide-template* slab structure

B6 · Hidden Markov model (HMM) post-lodgement extension

If anyone asks about future methodological directions: - Statistician collaborator (Martin) has proposed a state-space / HMM approach - Treats latent population N_t as a hidden state under a stationary kernel - Aoristic-interval emission via Cox-process integration (baorista's existing infrastructure provides the emission layer) - Genuinely novel for archaeology (no prior art found in our literature scout) - Would be a post-lodgement OSF amendment if pursued - Citations: Kéry & Schaub 2012 IPM; Tingley & Huybers 2010 BARCAST